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1. Introduction

In this project, we ask whether machine learning tools improve portfolio construction or merely complicate the process without any value. So, we construct a Kelly-optimal portfolio of ten large US stocks, then we attempt three covariance-matrix techniques, including denoising, hierarchical clustering, and detoning, to see if any of them are worth it.

The results give a qualified answer: one technique was consistently helpful, one made no observable difference, and the simplest benchmark of all stayed competitive with everything else. We report the outcome exactly as we found it.

1.1 Data and windows

We chose the same ten securities as GWP1: TSLA, WMT, BAC, GS, LLY, MRK, GOOG, META, AAPL, and XOM. We work with daily adjusted closes over 502 trading days. The training window runs from 3rd January 2024 to 30 September 2025 (constituting 437 observations); the test window runs from 1 October to 31 December 2025 (64 observations). The two windows do not overlap. Step 3 uses the full collection of 501 return observations. We take the risk-free rate of 4.50% per annum.

Table 1: In-sample descriptive statistics

Ticker	Ann. mean (%)	Ann. volatility (%)	Skew	Excess kurtosis
TSLA	54.74	65.59	0.67	4.51
WMT	41.68	21.49	0.72	9.08
BAC	29.94	25.40	−0.64	8.78
GS	47.91	29.09	0.69	11.18
LLY	21.96	36.32	−0.35	8.43
MRK	−10.84	25.32	−0.88	6.89
GOOG	36.99	29.98	0.25	5.15
META	50.68	37.75	1.74	16.33
AAPL	22.90	29.13	1.00	13.19
XOM	11.49	21.89	−0.62	2.44

437 daily observations. Every asset has excess kurtosis ranging between 2.4 for Exxon to 16.3 for Meta. Returns are not normal at all. This is crucial since Kelly maximizes log growth, which is highly sensitive to the left tail. We optimize straight from the return distribution without resorting to Gaussian approximation.

1.2 A note on the brief

Then, we compare the Kelly portfolio with the double-Kelly and double-Kelly portfolios. We interpret this as the fractional comparison: half-Kelly, Kelly, and double-Kelly. This leads to three portfolios.

2. Step 1: The Optimal Kelly Portfolio

2.1 What Kelly maximizes

The Kelly strategy selects the portfolio with the highest expected log wealth, which equals the highest long-run geometric growth rate (Kelly 1956):

$$\max g(w) = E[\ln(1 + w^T r)] \text{ subject to } \Sigma w_i = 1,\ 0 \le w_i \le u$$

There are two things that must be remembered. For the infinite horizon, Kelly will beat any other fixed-fraction strategy almost surely. In the finite-horizon case, it is aggressive: log utility implies a relative risk aversion of exactly 1, much lower than in practice.

The optimal portfolio is found via empirical maximization of the sample mean of $\ln(1 + w^T r_t)$. Taking into account the kurtosis from Table 1, the use of the Gaussian approximation would mean throwing away the tail behavior, on which log utility is very keen.

2.2 Is the capped problem feasible?

Yes, and the argument is computational, not numerical. Given the 20% cap, 1 ÷ 0.20 = 5, therefore, there need to be at least five actively weighted positions. The universe has ten names, so feasible points are always attainable (any five-name equal-weighting is fine). The feasible set is a non-empty compact convex polytope, and $\ln(1 + w^T r)$ is concave in w; therefore, the maximum exists and is achieved.

Table 2: Kelly portfolio weights

Ticker	Kelly (no cap)	Kelly (20% cap)	Status
TSLA	15.10%	20.00%	At cap
WMT	0.00%	20.00%	At cap
BAC	0.00%	0.00%	—
GS	39.06%	20.00%	At cap
LLY	0.00%	0.00%	—
MRK	0.00%	0.00%	—
GOOG	0.00%	20.00%	At cap
META	45.83%	20.00%	At cap
AAPL	0.00%	0.00%	—
XOM	0.00%	0.00%	—
Total	**100%**	**100%**	

Uncapped Kelly portfolio holds three names; the capped version holds exactly five, all at the ceiling. Growth rate falls from 57.98% to 53.67% a year once the 20% cap is imposed, a cost of 4.31 percentage points.

The uncapped solution contains only three out of ten assets. Meta at 45.8%, Goldman Sachs at 39.1%, and Tesla at 15.1% make up the whole portfolio. Seven assets receive no weight, including Walmart, which has the second-best return-to-volatility ratio in the universe.

This is what log utility does when it believes in the input data. Since Kelly is more aggressive than mean-variance optimization because of its preference for geometric growth and treatment of estimated means as known ones, the three winners received the most realized geometric growth in the training period, and the optimizer focused on that.

The capped solution is a pure corner. All five holdings are exactly at 20%. No interior point. This implies that the cap determines the structure, not the covariance structure. The consequence of that observation is in Section 6, where we see that denoising the covariance matrix makes no difference.

3. Step 2: Kelly, Half-Kelly, and Double-Kelly

Fractional Kelly scales the allocation by a factor f and parks the remainder at the risk-free rate:

$$r_{p,t}(f) = f \cdot (w^T r_t) + (1 - f) \cdot r_f$$

f = 0.5 is half-Kelly, f = 1 is full Kelly, and f = 2 is double-Kelly, which borrows at the risk-free rate an amount equal to the portfolio value.

Table 3: Out-of-sample performance, 1 October to 31 December 2025

Measure	Half-Kelly (f=0.5)	Kelly (f=1)	Double-Kelly (f=2)
Cumulative return	4.66%	8.07%	14.42%
Ann. growth rate	19.64%	35.72%	69.96%
Sharpe ratio	1.4562	1.4562	1.4562
Ann. volatility	9.61%	19.21%	38.42%
Max drawdown	−3.16%	−6.52%	−13.07%
95% VaR (daily)	−0.91%	−1.84%	−3.70%
95% cVaR (daily)	−1.23%	−2.47%	−4.96%

Based on the 20%-capped Kelly weights. The three Sharpe ratios are identical to the fourth decimal position, as shown in the discussion below.

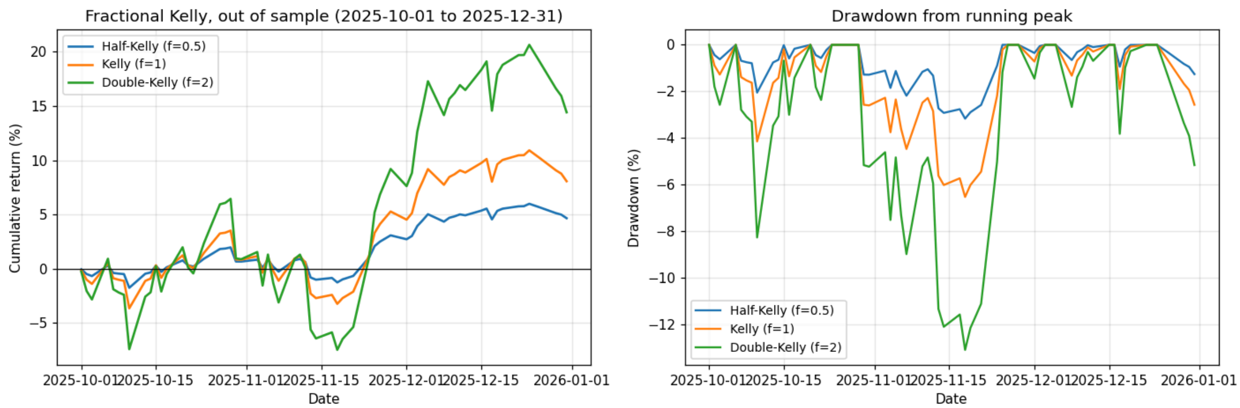

Figure 1: Fractional Kelly out-of-sample. Left: cumulative return. Right: drawdown from running peak. The three paths are the same shape scaled by f; only the vertical axis changes.

3.1 Why are the Sharpe ratios identical?

This is not a computational error, and it is the most useful thing in Table 3. Excess return scales exactly with the Kelly fraction:

$$r_{\square,\square}(f) - r\_f = f \cdot (w^T r_{\square} - r\_f)$$

Both the mean and the standard deviation of excess return therefore scale by a factor of f, ensuring the ratio between them is unchanged. **A Sharpe ratio cannot distinguish between Kelly fractions.** Any leverage decision judged solely on Sharpe alone is being judged on a statistic that is blind to it.

What separates them is presented in the remainder of the table. Doubling the fraction doubled the volatility from 9.6% to 38.4%, doubled the drawdown from -3.2% to -13.1%, and quadrupled the cVaR loss from -1.2% to -5.0%. The investor's actual experience differs enormously across the three; the Sharpe ratio records none of it.

3.2 Did double-Kelly win?

Over this quarter, yes, with 14.42% against 8.07%. But the quarter was strongly positive, and leverage always wins in a rising market. The cross-validation in the next section gives a fairer test, and there the conclusion reverses, especially in the weakest of the five folds, while double-Kelly grew at 6.12% a year against full Kelly's 7.38%.

That reversal is the whole point of the Kelly criterion. Growth rate is not monotonic in leverage. Past the optimal fraction, the drag from compounding losses outweighs the gain from compounding wins, and the growth rate falls. Fold 3 is that regime showing up in our data. Double-Kelly took roughly twice the drawdown for a lower growth rate, being the worst possible trade.

4. Step 3: Backtesting Via K-Fold Cross-Validation

4.1 Why traditional K-fold validation fails with asset returns.

Standard K-fold shuffles observations into folds, learns from K-1 partitions, and evaluates on the rest. Applied to a return series that is invalid for two reasons.

- **Leakage through time.** Shuffling puts future observations in the training set. A portfolio fitted on a dataset containing the test period will look better than it can possibly be in real time.
- **Leakage through serial correlation.** Even with contiguous folds, observations on either side of a boundary are associated with the evaluation dataset via volatility clustering. A model trained on day t-1 knows something about day t.

4.2 Our procedure

Following López de Prado (2018), we purge and embargo. Purging removes training observations that overlap the test window in terms of information, where the embargo removes a further block immediately after the test fold, because serial correlation runs forward as well as backward.

1. Split all 501 observations into five contiguous folds in chronological order, with no shuffling.
2. Loop through each fold as the test block, the remainder as training.
3. Purge and embargo five trading days on each side of the test block.
4. Re-solve the Kelly weights on the purged learning datasets.
5. Determine the untouched test fold for all three Kelly fractions.

The dispersion of results across folds is the measurement we care about. It estimates how much the Kelly weights are so dependent on the particular period used to fit them.

Table 4: Cross-validated performance by Kelly fraction

Fraction	Growth (mean)	Growth (sd)	Growth (min)	Sharpe (mean)	Sharpe (sd)	Max DD (mean)	cVaR (mean)
0.5	18.44%	7.30%	6.44%	1.3427	0.7061	−5.31%	−1.38%
1.0	33.25%	15.79%	7.38%	1.3427	0.7061	−10.59%	−2.79%
2.0	64.82%	36.36%	6.12%	1.3427	0.7061	−20.47%	−5.59%

Five purged folds over the full two years. Note the minimum column: double-Kelly's worst fold (6.12%) is below full Kelly's worst (7.38%), despite double the average drawdown.

Table 5: Kelly weights re-estimated on each fold

Ticker	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Std. dev.
TSLA	20.0%	0.0%	0.0%	2.1%	20.0%	0.1061
WMT	20.0%	20.0%	20.0%	20.0%	20.0%	0.0000
BAC	19.4%	0.0%	0.0%	0.0%	14.4%	0.0944
GS	20.0%	20.0%	20.0%	20.0%	20.0%	0.0000

Ticker	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Std. dev.
LLY	0.0%	20.0%	20.0%	20.0%	1.1%	0.1065
MRK	0.0%	0.0%	0.0%	0.0%	0.0%	0.0000
GOOG	20.0%	20.0%	20.0%	20.0%	4.4%	0.0696
META	0.6%	20.0%	20.0%	17.9%	20.0%	0.0850
AAPL	0.0%	0.0%	0.0%	0.0%	0.0%	0.0000
XOM	0.0%	0.0%	0.0%	0.0%	0.0%	0.0000

Weights were re-estimated independently on each fold's training data. Five of ten names are stable; the other five swing across the full range from 0% to the 20% ceiling.

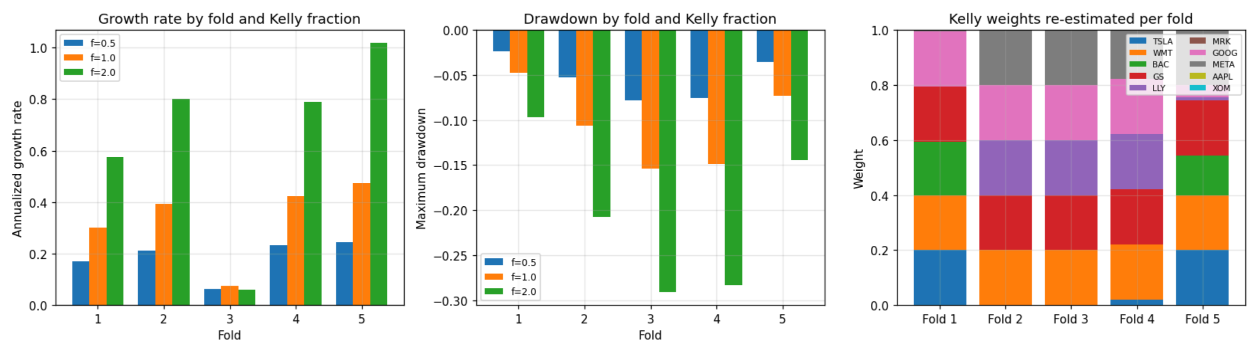

Figure 2: Cross-validation results. Left: growth rate by fold and Kelly fraction. Center: maximum drawdown. Right: Kelly weights are re-estimated per fold. Fold 3 is visibly the weakest period across all three panels.

4.3 What the folds reveal

The weights are unstable, and Table 5 shows exactly where. Walmart and Goldman Sachs hold 20% in every fold. Merck, Apple, and Exxon are excluded in every fold. Those five are a genuine signal, and the data agrees about them no matter which period it looks at.

The other five are noise. Tesla is at the 20% ceiling in folds 1 and 5 and at zero in folds 2 and 3. Eli Lilly is the mirror image: zero in fold 1, at the ceiling in folds 2 through 4. These names swing across the entire feasible range depending on which stretch of history the optimizer happens to see. Their standard deviations across folds, around 0.10, are half the size of the cap itself.

The performance dispersion is severe. The mean Sharpe across folds is 1.34 with a standard deviation of 0.71, so the dispersion is more than half the mean. Fold 3 returned a Sharpe of 0.24, while fold 5 returned 2.10. Any single-period backtest of this strategy could report almost any number in that range, and the number would tell you more about the period chosen than about the strategy.

This is exactly what the exercise is meant to expose. A single out-of-sample quarter is not evidence, and we treat the Step 6 results accordingly.

5. Step 4: Denoising, Clustering, and De-toning

5.1 Denoising

The problem is straightforward. A covariance matrix on N assets from T observations has $N(N+1)/2$ parameters from NT data points. When T/N is small, the sample eigenvalues spread out far more than the true ones, and most of that spread is pure noise. The damage shows up at inversion: an optimizer inverting Σ divides by the smallest eigenvalues, which are the ones most corrupted. Small spurious eigenvalues become large spurious weights.

The method follows Marchenko and Pastur (1967). They give the eigenvalue distribution of a purely random correlation matrix. For $q = T/N$ and noise variance σ^2, random eigenvalues fall between

$$\lambda_\pm = \sigma^2 \left(1 \pm \sqrt{(1/q)}\right)^2$$

Anything above λ_+ is treated as a signal. We fit σ^2 by minimizing the distance between the empirical eigenvalue density and the theoretical Marchenko-Pastur density, then replace every eigenvalue below λ_+ with their common average. This preserves the trace, so total variance is unchanged, and the matrix is rebuilt from the modified spectrum.

5.2 Clustering: Hierarchical Risk Parity

The problem. Optimizers invert the covariance matrix, and inversion needs the matrix to be well conditioned. Correlated assets make it ill-conditioned, and the optimizer responds by taking large offsetting positions in near-substitutes, which is precisely the behavior that fails out-of-sample. López de Prado (2016) calls this Markowitz's curse, whereby the more diversification is needed, the less reliable the optimizer becomes.

The method. HRP never inverts anything, though it converts correlations to distances, $d_{ij} = \sqrt{((1-\rho_{ij})/2)}$, builds a hierarchical tree, reorders the covariance matrix so similar assets sit adjacent, then splits recursively by allocating between each pair of sub-clusters in inverse proportion to their variance.

The practical benefit is that the weights are contingent upon the structure of the correlation matrix rather than its inverted form, which allows results to be stable and works even in cases where the matrix is singular.

5.3 De-toning

The problem. In equity data, the largest eigenvector is the market factor, and it dominates every pairwise correlation. A clustering algorithm examining unadjusted correlations mostly sees the market, not the structure underneath.

The method. After denoising, we isolate the difference from the first eigenvector, then rescale back to a correlation matrix, and therefore, what remains describes how assets move relative to the market rather than with it.

One important caution here is that detoning ensures that the matrix is singular. Omitting one eigenvector results in the rank equal to N−1; hence, the smallest eigenvalue will necessarily be zero. Inverting the detoned matrix via an allocation routine is not recommended. There is no such restriction for HRP, though, which does not invert.

5.4 Results on our data

Table 6: Denoising and detoning diagnostics

Quantity	Value	Reading
q = T / N	43.7	437 observations, 10 assets
Fitted noise variance σ^2	0.8611	From the MP density fit
MP cutoff λ_+	1.1413	Above this is signal
Eigenvalues retained	3 of 10	Seven judged to be noise
Largest eigenvalue	3.5508	The market factor
Smallest raw eigenvalue	0.2093	Most noise-corrupted
Denoised residual eigenvalue	0.5786	Common replacement value
Condition number, raw	16.96	—
Condition number, denoised	6.97	58.9% reduction
Condition number, detoned	3.72×10^{15}	Singular - do not invert
Market share of variance	34.7%	First eigenvector

The trace is held strictly constant at 10.0000 before and after denoising, which confirms that total variance is unchanged.

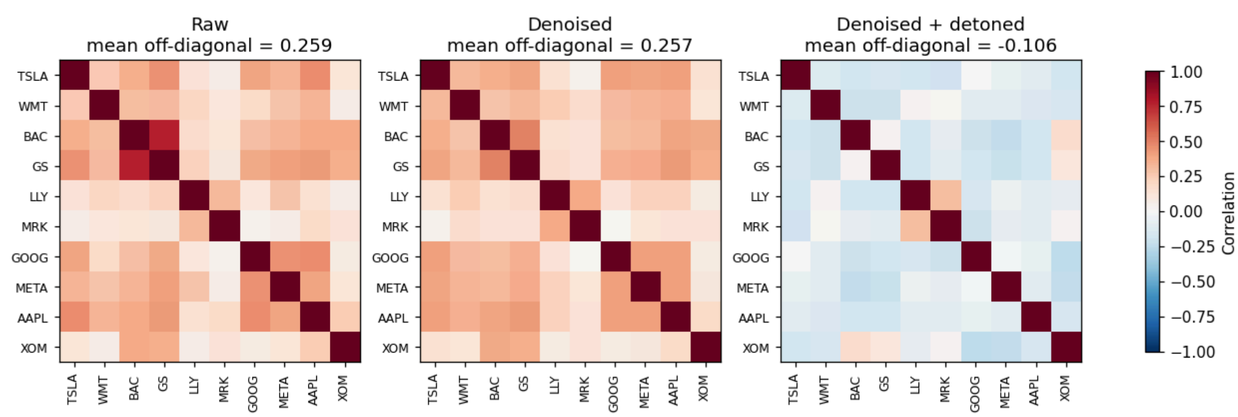

Figure 3 presents the correlation matrices under the three treatments. Denoising did not affect the mean off-diagonal correlation much, leaving it almost the same, 0.257, versus 0.259 prior to denoising; detoning decreased it to -0.106 by removing the market factor.

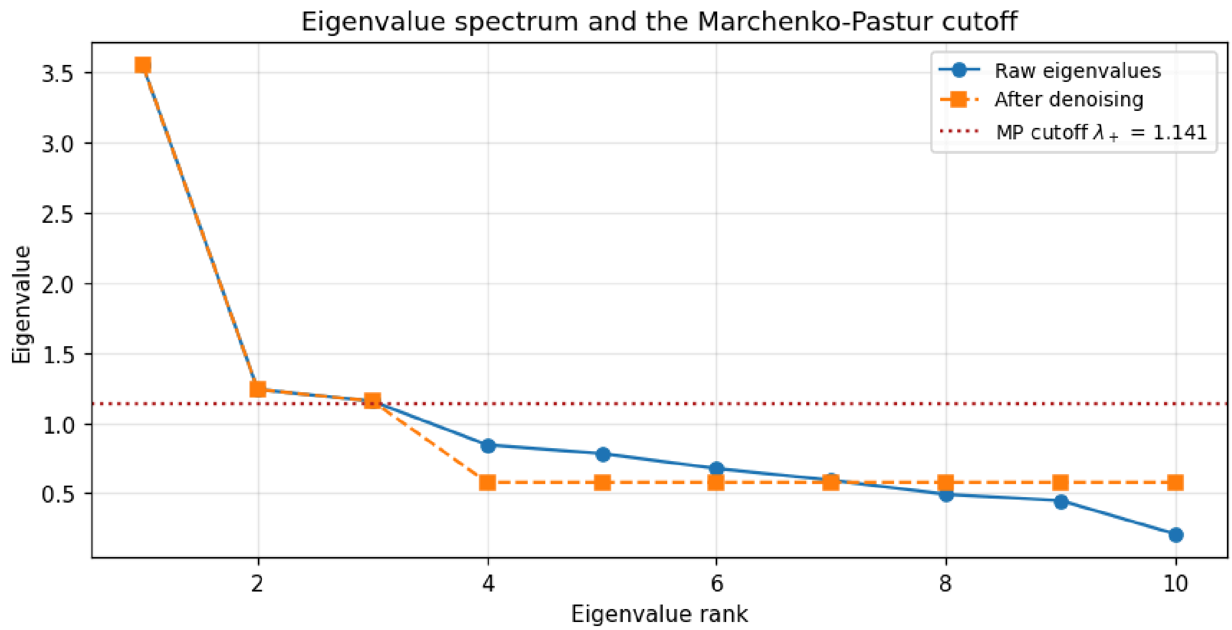

Figure 4 depicts the spectrum of eigenvalues compared to the Marchenko-Pastur cutoff. There are three distinct eigenvalues, $\lambda_+ = 1.1413$; the other seven are substituted by their common average, 0.5786.

Denoising succeeded to do its job. The condition number dropped from 16.96 to 6.97, that is, 58.9%, and the smallest eigenvalue increased 2.8 times from 0.209 to 0.579. Three out of ten eigenvalues are greater than the cutoff level; thus, seven of the ten possible directions of the asset correlation were considered by the procedure as pure noise.

Detoning did precisely what it was supposed to do in the theory. The mean off-diagonal correlation dropped from 0.257 to -0.106, so the market factor was indeed the most prominent shared component. The smallest eigenvalue turned out to be 4.7×10^{-16}, that is, zero up to machine precision; the condition number rose to 3.7×10^{15}. We confirmed this property numerically rather than assumed, and this is why each detoned portfolio in the present report is computed via HRP.

Table 7: HRP weights under three covariance treatments

Ticker	HRP (raw)	HRP (denoised)	HRP (den. + detoned)
TSLA	2.23%	1.81%	2.35%
WMT	20.18%	20.90%	16.83%
BAC	6.55%	7.58%	12.82%
GS	4.99%	5.77%	8.06%
LLY	6.04%	6.85%	4.64%
MRK	12.43%	14.09%	10.64%
GOOG	10.69%	10.06%	11.23%
META	5.24%	5.48%	7.04%
AAPL	12.21%	8.59%	9.16%

Ticker	HRP (raw)	HRP (denoised)	HRP (den. + detoned)
XOM	19.44%	18.86%	17.24%
Effective N	**7.47**	**7.48**	**8.27**

Effective N is 1 divided by the sum of squared weights. All three HRP portfolios hold all ten names, against five for Kelly.

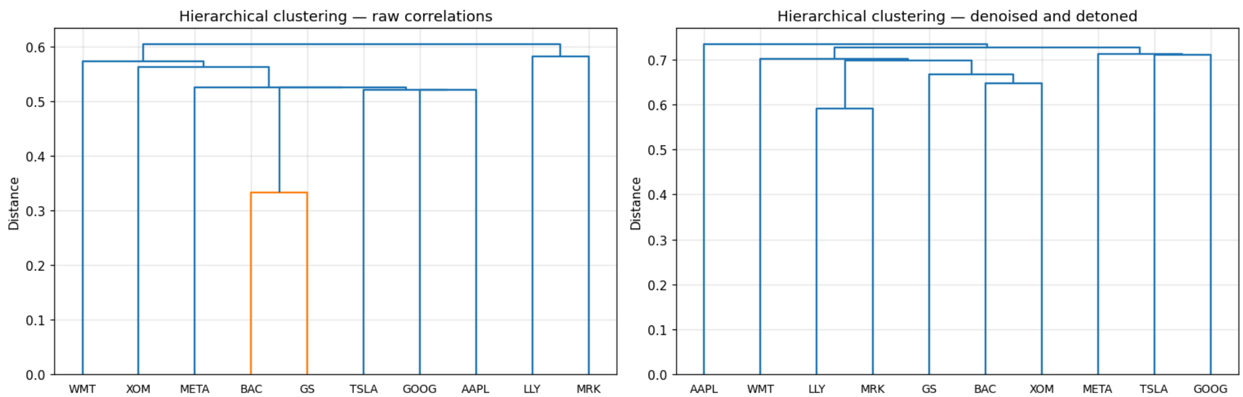

Figure 5: Hierarchical clustering dendrograms. Left: raw correlations. Right: denoised and detoned. Removing the market factor changes the cluster ordering substantially; the banks (BAC, GS) pair more tightly once market co-movement is stripped out.

Detoning changed the clustering, which is exactly what it is meant to do. The raw ordering starts with WMT, XOM, and META; the detoned ordering starts with AAPL, WMT, and LLY. More usefully, the effective number of positions rose from 7.47 to 8.27, so the detoned portfolio is genuinely better spread. And, with the market factor removed, the algorithm can see relative structure it previously could not.

One honest problem remains. HRP raw and HRP denoised put 20.18% and 20.90% into Walmart, breaching the 20% cap imposed in Step 1. HRP takes no constraints, but it allocates by hierarchy, and nothing in the algorithm knows about a position limit. Only the detoned version comes in under the cap, at 17.24%, and that is coincidence rather than design. A live mandate with a hard 20% limit would need the HRP output post-processed, and we flag the issue rather than quietly rescaling the numbers.

6. Steps 5 and 6: Which Combination Wins?

6.1 How the improvements can be sequenced

The three tools act on different objects, so they cannot be stacked in any order we like:

- **Denoising** modifies the covariance matrix and helps anything that inverts it.
- **Detoning** modifies it further and prevents inversion entirely.
- **HRP** replaces the optimizer and never inverts.

So denoising must therefore come before detoning, as detoning removes the leading eigenvector, and denoising needs the full spectrum to fit the cutoff. Detoned matrices can only be paired with HRP. To give denoising any chance to act on a Kelly allocation at all, we switch to the second-order approximation

$g(w) \approx w^T\mu - ½w^T\Sigma w$, because the empirical Kelly of Step 1 has no covariance matrix for these tools to improve.

Equal weight is included as a benchmark. DeMiguel, Garlappi, and Uppal (2009) found 1/N beats most optimized allocations out of sample, so any improvement that cannot beat it has not justified its complexity.

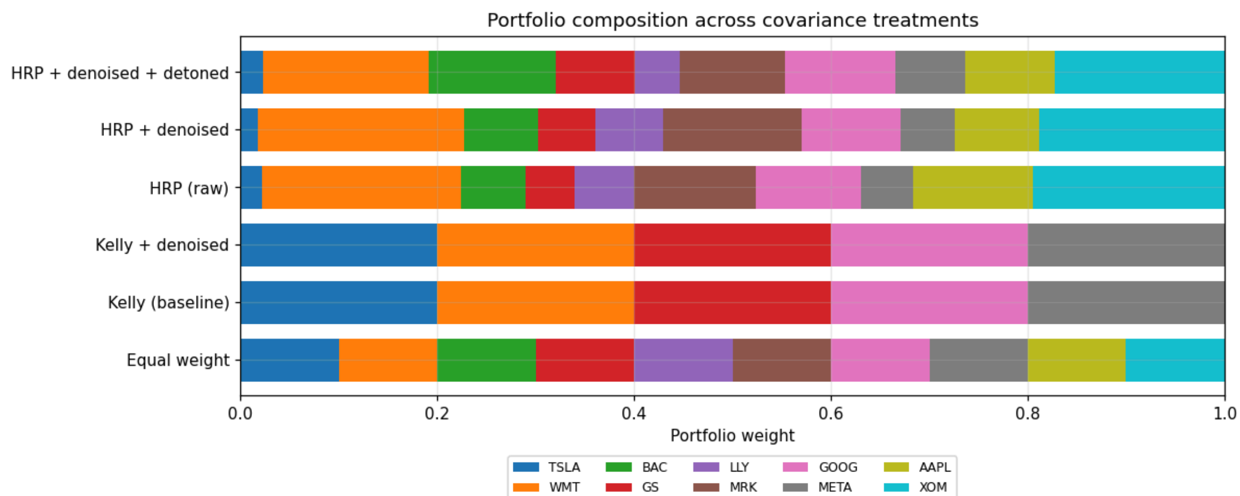

Figure 6 shows portfolio composition across treatments. The two Kelly portfolios hold five names at 20% each; the three HRP portfolios hold all ten.

6.2 Out-of-sample results

Table 8: Out-of-sample results, 1 October to 31 December 2025

Measure	Equal wt.	Kelly	Kelly + den.	HRP raw	HRP + den.	HRP+den+det
Cumulative return	12.89%	8.07%	8.07%	13.50%	13.97%	12.56%
Ann. growth rate	61.19%	35.72%	35.72%	64.63%	67.35%	59.32%
Sharpe ratio	3.3215	1.4562	1.4562	4.3115	4.4216	3.8715
Sortino ratio	5.2283	2.2336	2.2336	6.8547	7.0584	6.0215
Ann. volatility	13.32%	19.21%	19.21%	10.68%	10.79%	11.06%
Max drawdown	−3.61%	−6.52%	−6.52%	−3.07%	−2.96%	−3.17%
95% cVaR (daily)	−1.51%	−2.47%	−2.47%	−1.26%	−1.26%	−1.31%

Six metrics across all six portfolios. Note that Kelly and Kelly + denoised are identical in every column.

Table 9: Average rank across the six metrics (1 = best)

Portfolio	Cum. ret.	Growth	Sharpe	Sortino	Max DD	cVaR	Avg rank
HRP + denoised	1	1	1	1	1	2	1.17
HRP (raw)	2	2	2	2	2	1	1.83
HRP + den. + detoned	4	4	3	3	3	3	3.33

Portfolio	Cum. ret.	Growth	Sharpe	Sortino	Max DD	cVaR	Avg rank
Equal weight	3	3	4	4	4	4	3.67
Kelly + denoised	5	5.5	5	5	5.5	6	5.33
Kelly (baseline)	6	5.5	6	6	5.5	5	5.67

HRP + denoised ranks first on five of six metrics on the test quarter.

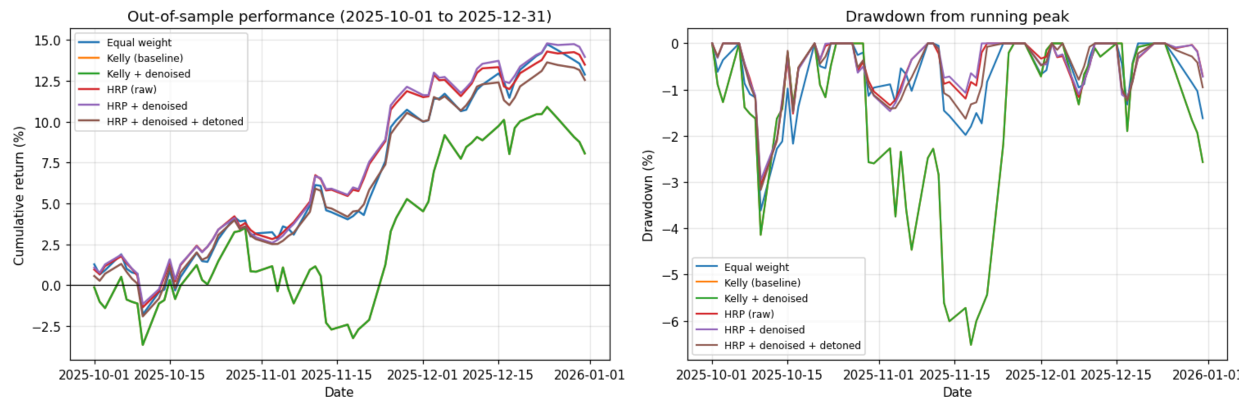

Figure 7: Out-of-sample performance and drawdown across all six portfolios. The three HRP variants cluster tightly together and above the two Kelly variants.

6.3 Question 1: What are the differences?

On the test quarter, the ordering is clear. HRP + denoised leads on five of six metrics with an average rank of 1.17, followed by HRP raw at 1.83. Both Kelly portfolios finish last.

The gap is large: HRP + denoised returned 13.97% against Kelly's 8.07%, at 10.79% volatility against 19.21%.

Two results deserve attention before anyone celebrates.

Denoising did nothing to the Kelly portfolio. Kelly and Kelly + denoised are identical to four decimal places in every column of Table 8, because their weights are identical. The reason is in Section 2.3: the capped Kelly solution is a pure corner, with all five positions at exactly 20%. The constraint fully determines the answer. Improving the covariance matrix cannot move a solution that is already pinned to the boundary of the feasible set.

Denoising barely moved HRP either. It improved the average rank from 1.83 to 1.17, but the underlying numbers moved by fractions of a percentage point, showing 13.97% against 13.50% cumulative return. The ranking overstates a difference that is small in economic terms.

6.4 The cross-validated check

A single 64-day quarter cannot establish that one method beats another. We therefore re-ran the whole comparison across the five purged folds, rebuilding every covariance matrix and every set of weights from each fold's training data.

Table 10: Cross-validated results across five purged folds

Portfolio	Growth (mean)	Growth (sd)	Sharpe (mean)	Sharpe (sd)	Max DD	cVaR
Equal weight	39.00%	27.97%	1.9936	1.7344	−9.70%	−2.43%
HRP (raw)	34.15%	28.63%	2.3182	2.3141	−7.49%	−1.92%
HRP + denoised	33.24%	27.31%	2.2144	2.2042	−7.66%	−1.95%
HRP + den. + detoned	35.58%	25.46%	2.2317	2.0079	−8.31%	−2.13%
Kelly (baseline)	33.25%	15.79%	1.3427	0.7061	−10.59%	−2.79%
Kelly + denoised	32.16%	14.76%	1.3049	0.6611	−10.67%	−2.79%

Every portfolio is rebuilt independently on each fold's training data. Standard deviations are reported because they are as informative as the means.

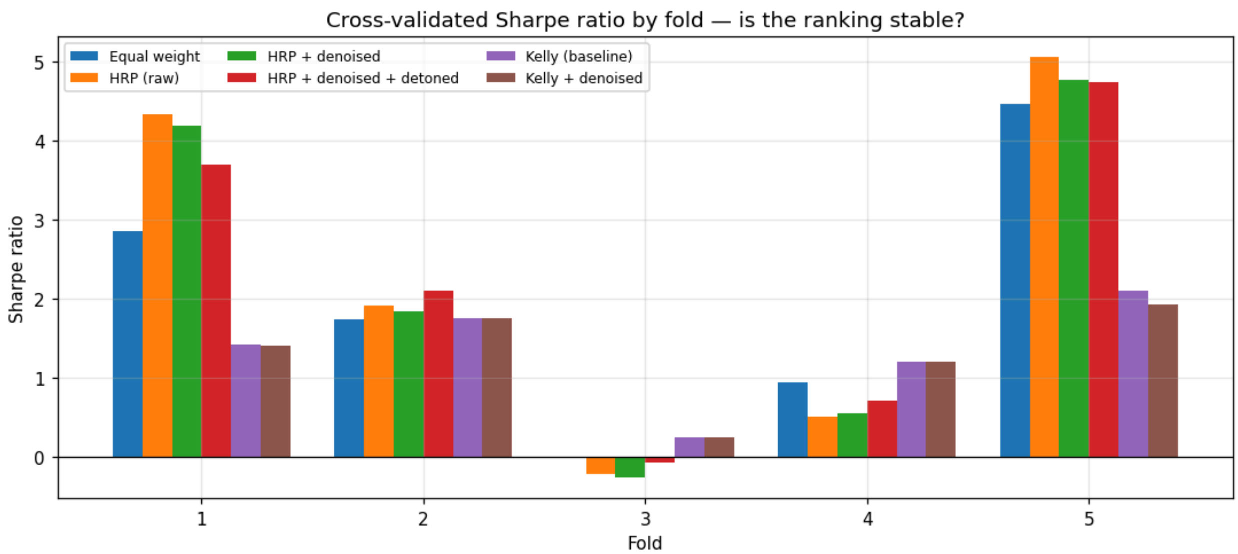

Figure 8: Cross-validated Sharpe ratio by fold. Fold 3 is negative for four of six portfolios. The dispersion across folds swamps the differences between portfolios.

The ranking does not survive. Equal weight posts the highest mean growth rate at 39.00%, ahead of every optimized alternative, including the HRP variants that won the test quarter. HRP keeps its edge on the risk metrics with a mean Sharpe of 2.32 against 1.99 and a mean drawdown of -7.49% against -9.70%, but its growth advantage disappears.

The dispersion is the real finding. HRP raw has a mean cross-validated Sharpe of 2.32 with a standard deviation of 2.31. The standard deviation equals the mean. In fold 3, four of the six portfolios posted negative Sharpe ratios. Against variation of that size, a difference of 0.1 in mean Sharpe is not a result, but it is noise with a decimal point.

While the two tests do not agree on growth, there is a commonality of opinion between the two regarding risk reduction, which holds more merit. HRP did better than every other strategy tested in terms of drawdown and conditional value at risk. This is what the evidence from our analysis proves.

6.5 Question 2: Why these differences?

q = 43.7 is the single best explanation for the poor denoising result. We have 437 observations for 10 assets, and the covariance was already well estimated, leaving not much noise to remove. Denoising is tailored to situations where N converges to T: hundreds of assets in a short window. Our problem is nowhere close to that regime, and the result is an honest signal about the value of the technique rather than a criticism of its efficacy.

The 20% cap already performs part of HRP's function. A cap that requires at least five positions is a diversification constraint, and Jagannathan and Ma (2003) show that weight limits act as shrinkage on the covariance estimate. Two solutions to the same problem should not be expected to produce additive results. HRP's gain over Kelly comes almost entirely from holding ten names rather than five, and part of that gain was already present through the constraint.

The corner solution rendered improvements from covariance adjustments ineffective. This is the most critical point to understand in the context of the entire project. When the optimizer hits the constraint hard, as it did in every position, the inputs cease to matter. The final weights depend solely on the constraint strength and the relative attraction of the objects. The entire work put into covariance matrix estimation is wasted. One should assess the restrictiveness of constraints well before attempting any serious covariance shrinkage

One quarter is one regime. Optimization methods that reduce model risk pay off when the process accounts for shifts. Our test window was an extended period of positive US large-cap equity performance, and none of the folds contained a regime switch. The tools did not have an opportunity to demonstrate their worth.

6.6 Question 3: Does the gain justify the complexity?

Taking the three techniques separately and comparing them to the tables:

HRP: Yes, it did. The method either ranked first or second in every metric on the test quarter and produced the best mean Sharpe ratio, maximum drawdown, and cVaR across folds. It did not require any matrix inversion, return forecasts, or position constraints to generate a non-concentrated portfolio. In addition, it is the only approach that can operate with a singular covariance matrix, which is a significant edge in practice. This particular implementation is perhaps forty lines of code long, making it a superior choice for the risk-averse investor. For the risk-controlled mandate, there is no alternative in this study.

Denoising: not at this q. It did not improve the Kelly portfolio at all, and the changes to HRP were within a couple of percentage points of the significance threshold. The constraint number indeed increased by 58.9%, so the machinery worked, but there was nothing to improve. We would consider it if the universe exceeded 50 names, or the estimation window was within one year. At q = 43.7, it is effort without measurable return.

Detoning: not on its own evidence here. It ranked third on the test quarter, behind both other HRP variants, and its mean growth advantage across folds is well inside the noise. It did raise the effective number of positions from 7.47 to 8.27, and it produced the only HRP portfolio that respected the 20%

cap. There is a case for it where the market factor genuinely swamps the clustering, but our data does not make that case, as the market explained 34.7% of the total dispersion, which provides important information but is not overwhelming.

And equal weight deserves a mention. It had the highest growth rate across all portfolios, including ML-enhanced versions, with no estimation or coding overhead. DeMiguel, Garlappi, and Uppal (2009) basically predicted this outcome, so any recommendation from the current study should outperform it, and none does, on growth.

7. Conclusion

Kelly criterion induces concentration, and the cap defines the allocation. The uncapped Kelly portfolio contained 3 out of 10 names, while the 20% cap increased the number of positions to 5, each sizeable at the 20% level. That corner solution reduced growth by 4.31 p.p. per annum and rendered covariance adjustments ineffective, which explains why denoising produced no improvement.

The Sharpe ratio metric was misleading in this context as a performance measure. Half-, full-, and double-Kelly portfolios had the same Sharpe ratio of 1.4562 because the excess return was scaled proportionally to the risk. What differed was the drawdown trajectory: -3.16% versus -13.07% and the conditional VaR: -1.23% versus -4.96%. With respect to risk, the double-Kelly version was significantly worse. In the worst-case cross-validation fold, it grew more slowly than the full-Kelly portfolio while taking a double drawdown, illustrating the young investor's maxim: the leveraged growth reduces the growth potential, which is precisely the point of double-Kelly under-betting.

The cross-validation procedure changed the perspective on the matter. On the test quarter, HRP + denoising was better in 5 out of 6 metrics. Across five purged folds, equal weight had the highest mean growth, and the Sharpe standard deviation of HRP was within the mean: 2.31 versus 2.32. The only metric that survived the validation process in a positive light was the reduction in downside risk. HRP cut the mean drawdown from -10.59% to -7.49% and the cVaR from -2.79% to -1.92%, which is the type of improvement we felt comfortable claiming.

The real-world application of the current set of insights is informed by the principle of diagnosis before treatment. Denoising was not useful at q = 43.7 because the number of observations greatly exceeded the number of assets, which meant that little noise was present to begin with. No amount of eigenvalue clipping could improve the solution once it hit the hard constraint. The most important question in this context was not what technique is universally best but what particular challenge the analyst is trying to overcome at a given moment. In our case, the constraint domination was the issue, and the covariance shrinkage was a distraction, and identifying that is worth more than applying every tool available.

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